

Proposed Solution to #5815 SSMJ

Solution proposed by Prakash Pant, The University of Vermont, Bardiya, Nepal.

Problem Proposed by Goran Conar, Varaždin, Croatia.

Statement of the Problem: Let $a, x_1, x_2, \dots, x_n$ be positive real numbers. Prove the following inequality:

$$\left(\sum_{i=1}^n \frac{a^{x_i}}{x_i} \right) \geq \left(\sum_{i=1}^n \frac{1}{x_i} \right) \cdot a^{\frac{n}{\sum_{i=1}^n \frac{1}{x_i}}}.$$

When does equality occur?

Solution of the Problem: Rewrite the problem as :

$$\frac{\sum_{i=1}^n \frac{1}{x_i} a^{x_i}}{\sum_{i=1}^n \frac{1}{x_i}} \geq a^{\frac{n}{\sum_{i=1}^n \frac{1}{x_i}}}$$

Consider function $f(x) = a^x$ and weights $w(x) = \frac{1}{x}$. Notice that $f(x) = a^x$ is convex whenever x is positive as $f''(x) = a^x \ln^2(x)$ is positive for positive x . Thus, using Jensen inequality, we can say that

$$\frac{\sum_{i=1}^n w_i f(x_i)}{\sum_{i=1}^n w_i} \geq f\left(\frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}\right)$$

Using $w_i = \frac{1}{x_i}$ and $f(x) = a^x$ gives us our required result.

The equality occurs when $x_1 = x_2 = \dots = x_n$.